

Abelian Stability of the Period-4 Morton–Vivaldi Fibers

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Abstract

For a nontrivial root of unity ζ , put $K_\zeta = \mathbf{Q}(\zeta)$ and

$$F_\zeta(T) = T^6 + 2T^5 + 4T^4 + 6T^3 + (\zeta - 5)T^2 - 8T - 16.$$

We strengthen the period-4 Morton–Vivaldi irreducibility theorem by proving that F_ζ remains irreducible over the maximal abelian extension K_ζ^{ab} . Equivalently, if α is a root, then

$$K_\zeta(\alpha) \cap K_\zeta^{\text{ab}} = K_\zeta.$$

It follows that, over every finite abelian extension of $\mathbf{Q}$, the factorization of the period-4 pullback is caused entirely by the cyclotomic polynomial: there is no additional splitting inside a degree-six fiber. We also determine the maximal abelian subfield of each exact-period-4 parabolic parameter field.

1 Introduction

The companion paper [Wei26] proves the irreducibility of the period-4 cyclotomic pullbacks over $\mathbf{Q}$ and, through the multiplier-curve normalization, settles the complete period-4 case of the Morton–Vivaldi irreducibility conjecture. The present paper asks a stronger field-theoretic question about each degree-six fiber: can it split after an arbitrary abelian extension of its cyclotomic coefficient field? Theorem 8.1 answers this question negatively.

The proof has two local ingredients at primes above 2. First, the fiber F_ζ has an irreducible quartic factor over a completion of $\mathbf{Q}(\zeta)$. Second, an explicit $3+3$ resolvent has no root in $\mathbf{Q}(\zeta)$. An abstract criterion for irreducible sextics then excludes every possible nontrivial abelian subfield of a root field. The displayed resolvent and the three congruences used in its local analysis are certified by deterministic exact-integer computation.

The next sections develop the Newton-polygon input, the local quartic factor, the abstract sextic criterion, and the explicit resolvent calculation. The abelian-stability theorem then follows by combining these ingredients. The remaining sections describe factorization over abelian base fields, transfer the result to the period-4 delta factors, determine maximal abelian subfields of the corresponding parameter fields, and place the result in the context of torsion Hilbert irreducibility.

2 Set-up and established inputs

Define

$$g(T) = -T^4 - 2T^3 - 4T^2 - 6T + 5 + \frac{8}{T} + \frac{16}{T^2}.$$

For every root of unity ζ one has

$$g(T) - \zeta = -\frac{F_\zeta(T)}{T^2}. \quad (1)$$

The companion period-4 paper proves that

$$F_\zeta \text{ is irreducible over } \mathbf{Q}(\zeta) \quad (\zeta \neq 1). \quad (2)$$

We shall also use the local facts proved there at primes above 2: odd-order roots of unity reduce injectively, 2-power roots of unity reduce to 1, and the ramification index in the 2-power cyclotomic part has the values recalled below. In the case of a nontrivial odd part, the exact reduction and special-value valuation used in (3) were proved in the base irreducibility argument. No local factor of degree four was asserted there; that new conclusion is proved here. These established inputs are recorded in [Wei26].

3 Newton polygons and a denominator lemma

We first record precisely the local consequence of Newton polygons that will be used below. If L is a complete discretely valued field, its valuation is normalized by $v(L^\times) = \mathbf{Z}$, and the same letter v denotes a fixed extension to $\bar{L}$. We use the standard Newton polygon theorem over complete discretely valued fields [Ser79, Chapter II].

Lemma 3.1 (Newton-polygon denominator lemma). *Let $P \in L[T]$ be monic with $P(0) \neq 0$. Suppose that the Newton polygon of P has an edge of horizontal length ℓ and slope*

$$-\frac{h}{d}, \quad h \in \mathbf{Z}, \quad d \geq 1, \quad \gcd(h, d) = 1.$$

Then the slope factor belonging to that edge has degree ℓ , and every irreducible factor of the slope factor has degree divisible by d . In particular, if $\ell = d$, the slope factor is irreducible of degree d over L .

Proof. By the Newton polygon theorem over the complete, hence henselian, field L , the polynomial P has a factor $P_{h/d} \in L[T]$ whose roots, counted with multiplicity, are exactly the ℓ roots β of P for which

$$v(\beta) = \frac{h}{d}.$$

Thus $\deg P_{h/d} = \ell$.

Let $G \in L[T]$ be an irreducible factor of $P_{h/d}$ and let β be a root of G . Put $M = L(\beta)$ and let $e(M/L)$ be the ramification index. With the normalization $v(L^\times) = \mathbf{Z}$, the value group of M is

$$v(M^\times) = \frac{1}{e(M/L)}\mathbf{Z}.$$

Since $v(\beta) = h/d$ is in lowest terms, it follows that $d \mid e(M/L)$. The ramification index divides $[M : L] = \deg G$, so $d \mid \deg G$. If $\ell = d$, the degrees of the irreducible factors of $P_{h/d}$ are positive multiples of d whose sum is d ; hence there is exactly one such factor. $\square$

4 A local quartic factor of the period-4 fiber

Let $\zeta \neq 1$ be a root of unity, put $K = \mathbf{Q}(\zeta)$, and write

$$\text{ord}(\zeta) = 2^a m, \quad m \text{ odd}, \quad \zeta = \xi \eta,$$

where ξ has 2-power order and η has odd order. Recall that

$$F_\zeta(T) = T^6 + 2T^5 + 4T^4 + 6T^3 + (\zeta - 5)T^2 - 8T - 16.$$

Fix a prime $\mathfrak{p}$ of K above 2, write $K_{\mathfrak{p}}$ for the completion, normalize $v = v_{\mathfrak{p}}$ by $v(K_{\mathfrak{p}}^\times) = \mathbf{Z}$, and put $e = v(2)$.

The proof in the case $m > 1$ uses the following two local statements, which were established in the companion paper [Wei26]. Choose an odd-order root of unity $\theta \in K$ such that $\theta^4 = \eta$, and put $t_0 = \theta - 1$. Then t_0 is a $\mathfrak{p}$ -adic unit and

$$\overline{F_\zeta(T)} = T^2(T - \overline{t_0})^4, \quad v(F_\zeta(t_0)) = 1. \quad (3)$$

These are the only previously established local fiber statements used in the odd-part argument below.

Proposition 4.1 (Local quartic factor). *For every root of unity $\zeta \neq 1$, the polynomial F_ζ has an irreducible factor of degree 4 over $K_{\mathfrak{p}}$.*

Proof. Suppose first that $m > 1$. Write

$$F_\zeta(t_0 + Y) = \sum_{j=0}^6 b_j Y^j.$$

The second assertion in (3) gives $v(b_0) = 1$. Reducing the shifted polynomial and using the first assertion in (3), we obtain, in residue characteristic 2,

$$\begin{aligned} \overline{F_\zeta(t_0 + Y)} &= (Y + \overline{t_0})^2 Y^4 \\ &= Y^6 + \overline{t_0}^2 Y^4. \end{aligned}$$

Consequently

$$v(b_j) \geq 1 \quad (1 \leq j \leq 3), \quad v(b_4) = 0, \quad v(b_5) \geq 1, \quad v(b_6) = 0.$$

Every point $(j, v(b_j))$ lies on or above the line $y = 1 - j/4$, and equality holds at $j = 0$ and $j = 4$. Hence the Newton polygon contains the edge

$$(0, 1) \longrightarrow (4, 0),$$

of length 4 and slope $-1/4$. Lemma 3.1 shows that the corresponding degree-four slope factor of $F_\zeta(t_0 + Y)$ is irreducible over $K_{\mathfrak{p}}$. Translating $Y = T - t_0$ preserves degrees and irreducibility, and therefore gives the required factor of F_ζ .

Now suppose that $m = 1$. Since $\zeta \neq 1$, its order is 2^a with $a \geq 1$. The recalled local cyclotomic facts give

$$e = v(2) = 2^{a-1}.$$

For $a = 1$ one has $\zeta = -1$ and $v(\zeta - 5) = v(-6) = 1$. For $a \geq 2$, the element $\zeta - 1$ is a uniformizer, and

$$\zeta - 5 = (\zeta - 1) - 4$$

is a sum of terms of respective valuations 1 and $2e > 1$; hence again $v(\zeta - 5) = 1$. It follows directly from the seven coefficients of F_ζ , listed from the constant coefficient to the leading coefficient, that their valuations are

$$(4e, 3e, 1, e, 2e, e, 0). \quad (4)$$

The lower convex hull of these seven points is

$$(0, 4e) \longrightarrow (2, 1) \longrightarrow (6, 0). \quad (5)$$

Indeed, the point $(1, 3e)$ lies strictly above the first segment because

$$3e > \frac{4e + 1}{2},$$

while the extension of that segment has negative ordinate at every integer abscissa $j \geq 3$, since at $j = 3$ its ordinate is $-2e + 3/2 < 0$. All the remaining coefficient valuations are nonnegative. On the other hand, the points with abscissas 0 and 1 lie strictly above the line through $(2, 1)$ and $(6, 0)$, whose ordinates there are $3/2$ and $5/4$, respectively, because $4e > 3/2$ and $3e > 5/4$. Finally, the points with abscissas 3, 4, 5 lie strictly above the second segment, whose ordinates there are $3/4, 1/2, 1/4$, respectively. The two displayed slopes are

$$-\frac{4e - 1}{2} \quad \text{and} \quad -\frac{1}{4},$$

and are strictly increasing. These observations also show directly that no omitted point lies below either segment. The second edge in (5) has length 4 and reduced slope $-1/4$. Lemma 3.1 therefore makes its slope factor an irreducible quartic over K_p . $\square$

5 A sextic criterion for abelian disjointness

Let k be a number field and let

$$f(T) = \prod_{i=1}^6 (T - r_i) \in k[T]$$

be an irreducible separable sextic. For each unordered partition $\{A, A^c\}$ of $\{1, \dots, 6\}$ into two triples, put

$$s_A = \prod_{i \in A} r_i + \prod_{j \in A^c} r_j.$$

There are $\binom{6}{3}/2 = 10$ such partitions. Define the $3 + 3$ resolvent by

$$\mathcal{R}_f(S) = \prod_{\{A, A^c\}} (S - s_A). \quad (6)$$

The absolute Galois group of k permutes the ten unordered partitions; thus $\mathcal{R}_f(S) \in k[S]$.

Write k^{ab} for the maximal abelian extension of k in a fixed algebraic closure.

Theorem 5.1 (Abelian-disjointness criterion for sextics). *Let α be a root of f and put $E = k(\alpha)$. Suppose that*

- (i) *for some finite place v of k , the polynomial f has an irreducible quartic factor over k_v ;*

(ii) the resolvent $\mathcal{R}_f$ has no root in k .

Then

$$E \cap k^{\text{ab}} = k.$$

Consequently f remains irreducible over k^{ab} .

Proof. Set

$$D = E \cap k^{\text{ab}}.$$

Because $D \subseteq E$, the extension D/k is finite. Moreover, k^{ab}/k is Galois with abelian Galois group. Every finite intermediate extension is therefore Galois and abelian over k ; in particular, so is D/k . Since f is irreducible of degree six, $[E : k] = 6$, and the tower $k \subseteq D \subseteq E$ gives

$$[D : k] \mid 6.$$

It remains to exclude the three possibilities $[D : k] = 2, 3, 6$.

Suppose first that $[D : k] = 2$, and let τ be the nontrivial element of $\text{Gal}(D/k)$. The minimal polynomial H of α over D is monic of degree $[E : D] = 3$ and divides f in $D[T]$. The polynomials H and $\tau(H)$ are distinct: otherwise H would have coefficients fixed by τ , hence would lie in $k[T]$, contradicting the irreducibility of the degree-six polynomial f over k . Both are irreducible over D , so they are coprime. Their product is monic of degree six, divides f , and is fixed by τ ; therefore

$$f = H \tau(H).$$

The roots of H form a triple $\{r_i : i \in A\}$ and those of $\tau(H)$ form the complementary triple. Thus

$$H(0) = - \prod_{i \in A} r_i, \quad \tau(H)(0) = - \prod_{j \in A^c} r_j,$$

and consequently

$$- \text{Tr}_{D/k}(H(0)) = \prod_{i \in A} r_i + \prod_{j \in A^c} r_j = s_A.$$

This is a root of $\mathcal{R}_f$ lying in k , contrary to hypothesis (ii). Hence $[D : k] \neq 2$.

Suppose next that $[D : k] = 3$. Let $q \in k_v[T]$ be an irreducible quartic factor of f , let β be a root of q , and choose a k -embedding

$$\sigma : E \hookrightarrow \overline{k_v} \quad \text{with} \quad \sigma(\alpha) = \beta.$$

Such an embedding exists because β is a root of the minimal polynomial f of α . Put $L = k_v(\beta)$, so $[L : k_v] = 4$. The compositum $k_v\sigma(D) \subset \overline{k_v}$ is canonically isomorphic to the completion D_w at the place w induced by σ , and it is contained in L because $D \subseteq E = k(\alpha)$.

Since D/k is Galois of prime degree three, the local degree $[D_w : k_v]$ is either 1 or 3. If it is 3, then the tower

$$k_v \subseteq k_v\sigma(D) \subseteq L$$

would force 3 to divide $[L : k_v] = 4$, which is impossible. If it is 1, then v splits completely in D and $\sigma(D) \subseteq k_v$. The minimal polynomial of α over D has degree $[E : D] = 2$; applying σ to its coefficients gives a quadratic in $k_v[T]$ having β as a root. This implies $[k_v(\beta) : k_v] \leq 2$, again contradicting $[L : k_v] = 4$. Thus $[D : k] \neq 3$.

Finally, suppose that $[D : k] = 6$. The inclusions $D \subseteq E$ and the equality of degrees give $D = E$. In particular, E/k is abelian Galois. With q , β , and σ as in the preceding paragraph, we have

$$k_v(\beta) = k_v\sigma(E),$$

because $E = k(\alpha)$. The field on the right is a completion of the Galois extension E/k , so its degree over k_v is the order of a decomposition group and therefore divides $[E : k] = 6$. Its degree is also 4, by the irreducibility of q . This contradiction excludes $[D : k] = 6$.

We have proved $D = k$. It remains to justify the assertion over the infinite extension k^{ab} . If f were reducible over k^{ab} , a nontrivial factorization would involve only finitely many coefficients. They would all lie in some finite intermediate extension

$$k \subseteq M \subseteq k^{\text{ab}}.$$

The extension M/k is finite abelian Galois, and

$$E \cap M \subseteq E \cap k^{\text{ab}} = k.$$

Thus $E \cap M = k$. Since M/k is Galois, the fields E and M are linearly disjoint over k . More explicitly, restriction identifies $\text{Gal}(EM/E)$ with $\text{Gal}(M/E \cap M) = \text{Gal}(M/k)$; hence $[EM : E] = [M : k]$, and the tower formula gives

$$[M(\alpha) : M] = [E : k] = 6.$$

The minimal polynomial of α over M therefore still has degree six and equals f , contradicting the assumed factorization over M . Hence f is irreducible over k^{ab} . $\square$

6 The 3+3 resolvent

Put

$$P_X(T) = T^6 + 2T^5 + 4T^4 + 6T^3 + (X - 5)T^2 - 8T - 16$$

so that $P_\zeta(T) = F_\zeta(T)$. Let $r_1, \dots, r_6$ be the roots of P_X in a splitting field over $\mathbf{Q}(X)$. If $A \subset \{1, \dots, 6\}$ has cardinality three, write $y_A = \prod_{i \in A} r_i$. For an unordered complementary pair $\{A, A^c\}$, put $s_A = y_A + y_{A^c}$. There are ten such pairs, and we define

$$\mathcal{R}_X(S) = \prod_{\{A, A^c\}} (S - s_A).$$

The coefficients are symmetric polynomials in the r_i with integer coefficients. The fundamental theorem on symmetric polynomials therefore gives $\mathcal{R}_X \in \mathbf{Z}[X, S]$.

Let C_X be the companion matrix of P_X , in a convention for which its characteristic polynomial is P_X , and put

$$Q_X(Y) = \det\left(YI - \bigwedge^3 C_X\right).$$

The eigenvalues of $\bigwedge^3 C_X$ are the twenty numbers y_A . Moreover,

$$y_A y_{A^c} = r_1 \cdots r_6 = -16, \quad y_{A^c} = -\frac{16}{y_A}.$$

Consequently, for each complementary pair,

$$(Y - y_A)(Y - y_{A^c}) = Y^2 - s_A Y - 16,$$

and hence

$$Q_X(Y) = Y^{10} \mathcal{R}_X \left(Y - \frac{16}{Y} \right). \quad (7)$$

Exact expansion gives

$$\begin{aligned} \mathcal{R}_X(S) = & S^{10} + 6S^9 + 4(X + 35)S^8 + 2(X^2 - 6X + 389)S^7 \\ & + (X^3 - 15X^2 + 667X + 5683)S^6 + 96(X^2 - 2X + 337)S^5 \\ & + 64(X^3 - 14X^2 + 477X + 800)S^4 - 1536(X^2 - 21X - 368)S^3 \\ & + 8192(X^2 + 35X - 48)S^2 - 65536(X - 16)(X + 4)S \\ & - 65536(X - 16)^2. \end{aligned} \quad (8)$$

Identity (8) is certified independently by the exact-arithmetic exterior-power computation described in Appendix A; no numerical specialization is used.

7 Absence of $\mathbf{K}$ -rational roots of the resolvent

Let $\zeta \neq 1$ be a root of unity, put $K = \mathbf{Q}(\zeta)$, and set $\mathcal{R}_\zeta = \mathcal{R}_X|_{X=\zeta}$. Fix a prime $\mathfrak{p} \mid 2$, normalize $v = v_{\mathfrak{p}}$ by $v(K_{\mathfrak{p}}^\times) = \mathbf{Z}$, and put $e = v(2)$. Write $\bar{x}$ for the reduction of an integral element x modulo $\mathfrak{p}$.

7.1 Orders with a nontrivial odd part

Write $\text{ord}(\zeta) = 2^a m$ with $m > 1$ odd, and decompose $\zeta = \xi\eta$, where ξ has 2-power order and η has odd order. Since $\bar{\xi} = 1$ and reduction is injective on odd-order roots of unity, $\bar{\zeta} = \bar{\eta} \neq 1$.

Write $\mathcal{R}_\zeta(S) = \sum_{j=0}^{10} c_j S^j$. Reducing the unit part of each coefficient in (8) gives the following exact valuations:

$$(v(c_0), \dots, v(c_{10})) = e(16, 16, 13, 9, 6, 5, 0, 1, 2, 1, 0).$$

For completeness, the reductions that prove exactness are

$$\begin{aligned} \left(c_j / 2^{v(c_j)/e} \bmod \mathfrak{p} \right)_{j=0}^{10} = & (\bar{\zeta}^2, \bar{\zeta}^2, \bar{\zeta}(\bar{\zeta} + 1), \bar{\zeta}(\bar{\zeta} + 1), \\ & \bar{\zeta}(\bar{\zeta} + 1)^2, (\bar{\zeta} + 1)^2, (\bar{\zeta} + 1)^3, (\bar{\zeta} + 1)^2, \\ & \bar{\zeta} + 1, 1, 1). \end{aligned}$$

Odd rational-integer factors reduce to 1 and have therefore been suppressed; every displayed residue is nonzero.

The lower convex hull is

$$(0, 16e) \longrightarrow (6, 0) \longrightarrow (10, 0). \quad (9)$$

Indeed, after division by e , the differences between the valuations at abscissas $1, \dots, 5$ and the line from $(0, 16e)$ to $(6, 0)$ are

$$\frac{8}{3}, \quad \frac{7}{3}, \quad 1, \quad \frac{2}{3}, \quad \frac{7}{3},$$

all positive. From abscissa 6 onward the lower boundary is horizontal at height 0. The first six roots thus have valuation $8e/3$. The ramification index e is a power of two, so $8e/3 \notin \mathbf{Z}$; none of these six roots belongs to $K_{\mathfrak{p}}$.

It remains to exclude the four unit roots. Since $\gcd(4, m) = 1$, the map $x \mapsto x^4$ is an automorphism of the cyclic group $\langle \eta \rangle$. Let $\theta \in \langle \eta \rangle$ be the unique element satisfying $\theta^4 = \eta$, and put

$$s_0 = (\theta + 1)^3.$$

Since $\theta \neq 1$ and reduction is injective on odd-order roots of unity, $\bar{\theta} \neq 1$; hence s_0 is a $\mathfrak{p}$ -adic unit. Only the coefficients of S^{10} and S^6 survive modulo $\mathfrak{p}$, and

$$\bar{c}_6 = (\bar{\zeta} + 1)^3 = (\bar{\theta} + 1)^{12} = \bar{s}_0^4.$$

Thus

$$\overline{\mathcal{R}_\zeta(S)} = S^6(S - \bar{s}_0)^4. \quad (10)$$

We shall use the following three coefficientwise congruences in $\mathbf{Z}[u]$:

$$\mathcal{R}_{u^4}((u+1)^3) \equiv 2u^3(u+1)^{27} \pmod{4}, \quad (11)$$

$$\left. \frac{\partial \mathcal{R}_X}{\partial X}((u+1)^3) \right|_{X=u^4} \equiv (u+1)^{26} \pmod{2}, \quad (12)$$

$$\mathcal{R}_{-u^4}((u+1)^3) \equiv 2u^3(u+1)^{26} \pmod{4}. \quad (13)$$

They are verified by the exact-arithmetic certificate described in Appendix A, which checks every coefficient of the three differences.

We now prove

$$v(\mathcal{R}_\zeta(s_0)) = 1. \quad (14)$$

If $a = 0$, then $\zeta = \eta = \theta^4$ and $e = 1$; congruence (11) writes $\mathcal{R}_\zeta(s_0)$ as twice a unit plus an element of $4\mathcal{O}_{K_{\mathfrak{p}}}$, and proves (14). If $a = 1$, then $\zeta = -\eta = -\theta^4$ and $e = 1$, so (13) gives the same conclusion in the same way.

Suppose $a \geq 2$. Set $\delta = \eta(\xi - 1)$, so that $\zeta = \eta + \delta$ and $v(\delta) = 1$. Taylor expansion in the polynomial variable X gives

$$\mathcal{R}_{\eta+\delta}(s_0) = \mathcal{R}_\eta(s_0) + \delta \partial_X \mathcal{R}_X(s_0)|_{X=\eta} + \delta^2 A$$

for some $A \in \mathcal{O}_{K_{\mathfrak{p}}}$; its integrality follows because $\mathcal{R}_X(s_0)$ has coefficients in $\mathcal{O}_{K_{\mathfrak{p}}}$. Congruence (11) gives $v(\mathcal{R}_\eta(s_0)) = e \geq 2$, while (12) says that the displayed derivative is a unit. The linear term therefore has valuation one and every other term has valuation at least two, proving (14) without cancellation.

After the translation $S = s_0 + Y$, all coefficients remain integral, and equation (10) shows that the coefficients of Y, Y^2, Y^3 have positive valuation and that the coefficient of Y^4 is a unit. Together with (14), this puts the edge

$$(0, 1) \longrightarrow (4, 0)$$

in the Newton polygon. Its slope is $-1/4$ and its horizontal length is four. Lemma 3.1 therefore makes the corresponding degree-four factor irreducible. In particular, none of these four roots lies in $K_{\mathfrak{p}}$. Together with (9), this proves that $\mathcal{R}_\zeta$ has no root in K whenever $m > 1$.

7.2 Pure powers of two

First let $\zeta = -1$. Substitution in (8) gives $c_j = 2^{v(c_j)}u_j$, with the following odd unit parts u_j :

j	0	1	2	3	4	5	6	7	8	9	10
u_j	-289	51	-41	519	77	255	625	99	17	3	1,

and therefore

$$(v(c_0), \dots, v(c_{10})) = (16, 16, 14, 10, 8, 7, 3, 3, 3, 1, 0).$$

The differences above the line from $(0, 16)$ to $(6, 3)$ at abscissas $1, \dots, 5$ are

$$\frac{13}{6}, \quad \frac{7}{3}, \quad \frac{1}{2}, \quad \frac{2}{3}, \quad \frac{11}{6}.$$

The points at abscissas $7, 8, 9$ also lie strictly above the line from $(6, 3)$ to $(10, 0)$. Hence the lower convex hull is

$$(0, 16) \longrightarrow (6, 3) \longrightarrow (10, 0),$$

with slopes $-13/6$ and $-3/4$. The six roots belonging to the first edge have valuation $13/6$, and the four belonging to the second have valuation $3/4$. Neither valuation is integral. In particular, $\mathcal{R}_{-1}$ has no root in $\mathbf{Q}_2$.

Now suppose that ζ has order 2^a with $a \geq 2$. Put $\pi = \zeta - 1$ and $e = v(2) = 2^{a-1}$, so $v(\pi) = 1$. Substituting $X = 1 + \pi$ into the parenthesized coefficient polynomials in (8) gives

$$\begin{aligned} X^2 + 35X - 48 &= \pi^2 + 37\pi - 12, \\ X^2 - 21X - 368 &= \pi^2 - 19\pi - 388, \\ X^3 - 14X^2 + 477X + 800 &= \pi^3 - 11\pi^2 + 452\pi + 1264, \\ X^2 - 2X + 337 &= \pi^2 + 336, \\ X^3 - 15X^2 + 667X + 5683 &= \pi^3 - 12\pi^2 + 640\pi + 6336, \\ X^2 - 6X + 389 &= \pi^2 - 4\pi + 384, \\ X + 35 &= \pi + 36. \end{aligned}$$

In addition, $X - 16 = \pi - 15$ and $X + 4 = \pi + 5$ are units. Since $e \geq 2$, each displayed expression has a unique term of least valuation. It follows that

$$(v(c_0), \dots, v(c_{10})) = (16e, 16e, 13e + 1, 9e + 1, 6e + 2, 5e + 2, 3, e + 2, 2e + 1, e, 0). \quad (15)$$

For the first segment, the differences at abscissas $1, \dots, 5$ are

$$\frac{16e - 3}{6}, \quad \frac{7e}{3}, \quad \frac{2e - 1}{2}, \quad \frac{2e}{3}, \quad \frac{14e - 3}{6},$$

all positive. At abscissas $7, 8, 9$, the valuations $e + 2, 2e + 1, e$ are strictly greater than the respective heights $9/4, 3/2, 3/4$ of the second segment. Thus the lower convex hull is

$$(0, 16e) \longrightarrow (6, 3) \longrightarrow (10, 0),$$

with slopes

$$-\frac{16e - 3}{6}, \quad -\frac{3}{4}.$$

The integer $16e - 3$ is odd, and modulo three it is congruent to e . Since e is a power of two, $3 \nmid e$; hence

$$\gcd(16e - 3, 6) = 1.$$

The two slopes have reduced denominators six and four, respectively, and the corresponding horizontal lengths are six and four. Lemma 3.1 therefore shows that the two slope factors are irreducible of degrees six and four. In particular, $\mathcal{R}_\zeta$ has no root in $K_{\mathfrak{p}}$.

Combining the odd-part and pure-power cases proves that

$$\mathcal{R}_\zeta(S) \text{ has no root in } K \quad (\zeta \neq 1). \quad (16)$$

8 Abelian stability

Theorem 8.1 (Abelian-stable irreducibility). *Let $\zeta \neq 1$ be any root of unity and put $K = \mathbf{Q}(\zeta)$. Then*

$$F_\zeta(T) \text{ is irreducible over } K^{\text{ab}}.$$

Equivalently, if α is a root of F_ζ , then

$$[K(\alpha) : K] = 6, \quad K(\alpha) \cap K^{\text{ab}} = K.$$

Proof. The base irreducibility statement (2) gives $[K(\alpha) : K] = 6$. Since K^{ab}/K is Galois, the standard degree formula gives

$$[K^{\text{ab}}(\alpha) : K^{\text{ab}}] = [K(\alpha) : K(\alpha) \cap K^{\text{ab}}].$$

Thus the stated intersection equality is equivalent to irreducibility over K^{ab} . Proposition 4.1 supplies an irreducible quartic local factor of F_ζ . Equation (16) proves that the $3 + 3$ resolvent has no root in K . Theorem 5.1 now gives both assertions. $\square$

9 Factorization over abelian base fields

Let Φ_k be the k -th cyclotomic polynomial, let φ be Euler's totient function, and define

$$\Gamma_k(T) = (-1)^{\varphi(k)} T^{2\varphi(k)} \Phi_k(g(T)).$$

For $k \geq 2$, the identity (1) gives

$$\Gamma_k(T) = \prod_{\substack{\zeta^k=1 \\ \zeta \text{ primitive}}} F_\zeta(T). \quad (17)$$

By Theorem 8.1, for every nontrivial root of unity ζ , the polynomial F_ζ is irreducible over $\mathbf{Q}(\zeta)^{\text{ab}}$, and for every root α one has

$$\mathbf{Q}(\zeta, \alpha) \cap \mathbf{Q}(\zeta)^{\text{ab}} = \mathbf{Q}(\zeta).$$

Fix a primitive k -th root ζ_k , put $K_k = \mathbf{Q}(\zeta_k)$, and let $B/\mathbf{Q}$ be a finite abelian extension. Set

$$D = B \cap K_k.$$

Because both $B/\mathbf{Q}$ and $K_k/\mathbf{Q}$ are Galois, restriction gives

$$\text{im}(\text{Gal}(\overline{\mathbf{Q}}/B) \longrightarrow \text{Gal}(K_k/\mathbf{Q})) = \text{Gal}(K_k/D).$$

The orbits on the primitive k -th roots therefore have cardinality $[K_k : D]$. Thus

$$\Phi_k(X) = \Psi_1(X) \cdots \Psi_r(X) \quad \text{in } B[X], \quad r = [D : \mathbf{Q}], \quad \deg \Psi_i = [K_k : D].$$

Here the Ψ_i are the distinct monic irreducible factors over B . For each i , define

$$\Gamma_{\Psi_i}(T) = (-1)^{\deg \Psi_i} T^{2 \deg \Psi_i} \Psi_i(g(T)).$$

The fixed-degree identity $T^2(g(T) - \zeta) = -F_\zeta(T)$ gives

$$\Gamma_{\Psi_i}(T) = \prod_{\Psi_i(\zeta)=0} F_\zeta(T) \quad \text{and} \quad \Gamma_k = \prod_{i=1}^r \Gamma_{\Psi_i}.$$

Fix a root ζ of Ψ_i and a root α of F_ζ . Since $F_\zeta(0) = -16$, one has $\alpha \neq 0$, so the Laurent expression $g(\alpha)$ is defined. The extension BK_k/K_k is abelian, hence is contained in K_k^{ab} . Therefore F_ζ remains irreducible over BK_k , and

$$[BK_k(\alpha) : BK_k] = 6.$$

On the other hand, the relation $\zeta = g(\alpha)$ shows that $BK_k \subset B(\alpha)$, and in fact $B(\alpha) = BK_k(\alpha)$. Hence

$$[B(\alpha) : B] = 6[BK_k : B] = 6[K_k : D].$$

This is the degree of Γ_{Ψ_i} . Since $\Gamma_{\Psi_i} \in B[T]$ is monic and vanishes at α , it is the minimal polynomial of α over B and is irreducible. These irreducible polynomials are pairwise distinct: if an element α were a common root of Γ_{Ψ_i} and Γ_{Ψ_j} , then $g(\alpha)$ would be a common root of the coprime polynomials Ψ_i and Ψ_j . This proves the following theorem.

Theorem 9.1. *For $k \geq 2$ and every finite abelian extension $B/\mathbf{Q}$, the polynomial Γ_k has exactly*

$$[B \cap K_k : \mathbf{Q}]$$

irreducible factors over B , each of degree

$$6[K_k : B \cap K_k].$$

In particular,

$$\Gamma_k \text{ is irreducible over } B \iff B \cap K_k = \mathbf{Q}.$$

All splitting after an abelian base change is therefore caused by the cyclotomic factor Φ_k ; no sextic fiber splits further.

Corollary 9.2 (Factorization over the maximal abelian extension). *For every $k \geq 2$, the complete irreducible factorization of Γ_k over $\mathbf{Q}^{\text{ab}}$ is*

$$\Gamma_k(T) = \prod_{\substack{\zeta^k=1 \\ \zeta \text{ primitive}}} F_\zeta(T). \quad (18)$$

It has $\varphi(k)$ factors, all of degree six.

Proof. Since $K_k \subset \mathbf{Q}^{\text{ab}}$ and $\mathbf{Q}^{\text{ab}}/K_k$ is abelian, it is contained in K_k^{ab} . Each F_ζ is consequently irreducible over $\mathbf{Q}^{\text{ab}}$. Distinct primitive roots give distinct polynomials, as their T^2 coefficients are $\zeta - 5$. Equation (17) now proves the assertion. $\square$

10 Transfer to the period-4 delta factors

We use the standard delta-factor notation [Mor96, MV95, KMST25]. Write $C = 4c$ for the scaled quadratic-family parameter and put

$$\tilde{\Delta}_{4k,4}(C) = \Delta_{4k,4}(C/4).$$

For a polynomial q , let $Z(q)$ denote its set of roots in $\overline{\mathbf{Q}}$, without multiplicity. The period-4 normalization has parameter coordinate

$$h(T) = -T^2 - 3 - \frac{4}{T}.$$

The established normalization and separability results [KMST25, Proposition 2.3, Equation (2.2), and Remark 2.10], together with the degree argument in [Wei26], show that the map $\beta(\alpha) = h(\alpha)$ supplies, for $k \geq 2$, a Galois-equivariant bijection between the root sets of the two separable polynomials,

$$\beta : Z(\Gamma_k) \longrightarrow Z(\tilde{\Delta}_{4k,4})$$

for $\text{Gal}(\overline{\mathbf{Q}}/\mathbf{Q})$. Restricting the action to $\text{Gal}(\overline{\mathbf{Q}}/B)$ preserves equivariance. The map β therefore identifies the Galois orbits over B , including their cardinalities. Since the irreducible factors of a separable polynomial correspond exactly to these orbits, Theorem 9.1 transfers without requiring an explicit polynomial substitution:

Theorem 10.1. *For $k \geq 2$ and every finite abelian extension $B/\mathbf{Q}$, the period-4 delta factor $\tilde{\Delta}_{4k,4}$ has exactly*

$$[B \cap K_k : \mathbf{Q}]$$

irreducible factors over B , each of degree

$$6[K_k : B \cap K_k].$$

Over $\mathbf{Q}^{\text{ab}}$ its complete factorization consists of $\varphi(k)$ irreducible sextic factors. Indeed, restricting β to $\text{Gal}(\overline{\mathbf{Q}}/\mathbf{Q}^{\text{ab}})$ still identifies the relevant orbits. More precisely, for every primitive k -th root ζ , the image under β of $Z(F_\zeta)$ is one orbit of cardinality six for $\text{Gal}(\overline{\mathbf{Q}}/\mathbf{Q}^{\text{ab}})$. The corresponding irreducible sextic delta factor is

$$H_\zeta(C) = \prod_{\alpha \in Z(F_\zeta)} (C - \beta(\alpha)) \in \mathbf{Q}^{\text{ab}}[C].$$

Remark 10.2. *This is an orbit description; it does not identify the polynomial $H_\zeta(C)$ in the parameter variable with $F_\zeta(T)$.*

For example, if $\gcd(k, \ell) = 1$, then $K_k \cap \mathbf{Q}(\zeta_\ell) = \mathbf{Q}$, and the period-4 delta factor remains irreducible over $\mathbf{Q}(\zeta_\ell)$.

11 The maximal abelian subfield of a parameter field

Corollary 11.1 (Maximal abelian subfield of a parameter field). *Let C be an exact-period-4 parabolic parameter with primitive k -th root multiplier ζ , where $k \geq 2$. Then*

$$\mathbf{Q}(C) \cap \mathbf{Q}^{\text{ab}} = \mathbf{Q}(\zeta).$$

Thus $\mathbf{Q}(\zeta)$ is the largest subfield of $\mathbf{Q}(C)$ that is Galois abelian over $\mathbf{Q}$, and $C \notin \mathbf{Q}^{\text{ab}}$.

Proof. Let α be the corresponding root of F_ζ , so $C = \beta(\alpha) = h(\alpha)$. The root correspondence is injective and $\text{Gal}(\overline{\mathbf{Q}}/\mathbf{Q})$ -equivariant. Consequently, for every $\sigma \in \text{Gal}(\overline{\mathbf{Q}}/\mathbf{Q})$,

$$\sigma(\alpha) = \alpha \iff \sigma(C) = C.$$

The two elements have the same stabilizer, and therefore

$$\mathbf{Q}(C) = \mathbf{Q}(\alpha).$$

Since $\zeta = g(\alpha)$, the right-hand field already contains $K = \mathbf{Q}(\zeta)$ and equals $K(\alpha)$. The abelian-stability theorem gives

$$K(\alpha) \cap K^{\text{ab}} = K.$$

Because $\mathbf{Q}^{\text{ab}}/K$ is abelian, $\mathbf{Q}^{\text{ab}} \subset K^{\text{ab}}$. Moreover, $K \subset \mathbf{Q}(C)$ and $K \subset \mathbf{Q}^{\text{ab}}$. The upper inclusion from the abelian-stability theorem and this lower inclusion give

$$\mathbf{Q}(C) \cap \mathbf{Q}^{\text{ab}} = \mathbf{Q}(\zeta). \quad (19)$$

Thus the multiplier field is precisely the largest subfield $L \subset \mathbf{Q}(C)$ for which $L/\mathbf{Q}$ is Galois abelian. In particular, $C \notin \mathbf{Q}^{\text{ab}}$: otherwise $\mathbf{Q}(C) \subset \mathbf{Q}^{\text{ab}}$ would force $\mathbf{Q}(C) = K$, contradicting $[\mathbf{Q}(C) : K] = 6$. $\square$

When the multiplier is 1, the exact-period-4 parameters instead satisfy

$$p(C) = C^3 + 9C^2 + 27C + 135 = 0.$$

After $U = C + 3$, this becomes $U^3 + 108$. A rational root of this monic cubic would be an integer r satisfying $r^3 = -108$, which is impossible because -108 is not an integer cube. The cubic is therefore irreducible. Its discriminant is

$$-27 \cdot 108^2 = -2^4 3^9,$$

which is not a square. Its Galois closure consequently has group S_3 , and the cubic field $\mathbf{Q}(C)$ is not Galois over $\mathbf{Q}$. Every finite subfield of $\mathbf{Q}^{\text{ab}}$ is Galois over $\mathbf{Q}$, so these parameters also lie outside $\mathbf{Q}^{\text{ab}}$. Since a cubic field has no nontrivial proper subfield, its largest subfield Galois abelian over $\mathbf{Q}$ is $\mathbf{Q}$. We conclude that no exact-period-4 parabolic parameter belongs to $\mathbf{Q}^{\text{ab}}$.

12 Torsion Hilbert irreducibility context

For this interpretation one must use the finite cover of degree six

$$g : U \longrightarrow \mathbf{G}_m, \quad U = \mathbf{G}_m \setminus g^{-1}(0).$$

Its degree is the degree of the rational function g , namely six. After writing the target coordinate as $X = g(T)$, the element T satisfies the monic equation $P_X(T) = 0$ with nonzero constant term; both T and T^{-1} are integral over $\overline{\mathbf{Q}}[X, X^{-1}]$, which also proves finiteness. The pullback condition means that the fiber product with $x \mapsto x^n$ is geometrically irreducible for every $n \geq 1$. Its function-field equation is

$$Y^n = g(T).$$

Over $\overline{\mathbf{Q}}(T)$, the function g is not a p -th power for any prime p . For $p \geq 3$, this follows from its pole of order two at $T = 0$. For $p = 2$, suppose $g = w^2$. Its only poles force

$$w = aT^2 + bT + c + dT^{-1}.$$

Comparison of the coefficients of T^4, T^3, T^{-1}, T^{-2} gives

$$a^2 = -1, \quad b = a, \quad d^2 = 16, \quad cd = 4.$$

Thus $c^2 = 1$, whereas comparison at T^2 gives

$$-1 + 2ac = -4, \quad ac = -\frac{3}{2},$$

so $(ac)^2 = 9/4$, contradicting $a^2c^2 = -1$. Hence g is not a square.

The field $\overline{\mathbf{Q}}(T)$ contains all roots of unity. The Kummer binomial criterion [Lan02] in this setting states that $Y^n - q$ is irreducible if and only if q is not a p -th power for every prime $p \mid n$. In the general binomial criterion one must also exclude $q \in -4\mathbf{Q}(T)^4$ when $4 \mid n$; here every element $-4b^4 = (2ib^2)^2$ is already a square, so the preceding square argument excludes this case as well. The preceding valuation and coefficient arguments therefore prove that $Y^n - g(T)$ is irreducible for every $n \geq 1$, which is precisely the pullback condition.

Zannier’s cyclotomic Hilbert irreducibility theorem [Zan10, Theorem 2.1] says that a cover defined over the cyclotomic closure and satisfying this condition has irreducible fibers over that field outside a finite union of proper torsion cosets. For $\mathbf{G}_m$, such a union is a finite set of torsion points. Thus the theorem gives finitely many exceptional torsion fibers. By the Kronecker–Weber theorem [Was97], the cyclotomic closure of $\mathbf{Q}$ is $\mathbf{Q}^{\text{ab}}$. The abelian-stability theorem determines them exactly: for $\zeta \neq 1$, the inclusion $\mathbf{Q}^{\text{ab}} \subset \mathbf{Q}(\zeta)^{\text{ab}}$ shows that F_ζ is irreducible over $\mathbf{Q}^{\text{ab}}$, whereas

$$F_1(T) = (T + 2)(T^2 + 4)(T^3 - 2).$$

Thus $\zeta = 1$ is the unique exceptional torsion fiber for this cover.

A Exact symbolic certificates

The verification directory contains an exact certificate using integer polynomial arithmetic and no third-party dependencies. It constructs the companion matrix of P_X , constructs the matrix of its third exterior power from all 3×3 minors, and verifies every coefficient of the exterior-power resolvent identity and the displayed resolvent formula. It also verifies the three polynomial congruences coefficient by coefficient. The program exits with a nonzero status on any failed equality or nonexact integer division; no floating-point computation or numerical specialization occurs.

From the directory containing this draft, the certificate is run with the command

```
python3 verification/verify_resolvent.py
```

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